👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.AT.8

Solve and Graph Inequalities

Lesson 7-6 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes
Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can solve an inequality and graph its solution set on a number line.

Language Objective: I can explain my work using the words solve, graph, solution, and substitute.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Solve | **Graph** | **Solution** | **Substitute** | **Solution set** | **Inverse operation**

👆 *Tap any word to see what it means and a picture.*

1. To find all the values that make an inequality true is to ________ it.
2. To show the solutions on a number line is to ________ them.
3. A value that makes an inequality true is a ________ .
4. To check a solution, I ________ the value back into the inequality.
5. All the values that make an inequality true form the ________ .
6. An operation that undoes another to isolate the variable is an ________ .

Reflect — Exit Ticket

Solve and describe the graph: $x + 5 < 11$

A. $x < 6$; open circle at 6, shade left

B. $x < 16$; open circle at 16, shade left

C. $x \leq 6$; closed circle at 6, shade left

D. $x > 6$; open circle at 6, shade right

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Solve
2. **Notes 2:** Graph
3. **Notes 3:** Solution
4. **Notes 4:** Substitute
5. **Notes 5:** Solution set
6. **Notes 6:** Inverse operation
7. **Watch for this mistake:** *A common mistake in Solve and Graph Inequalities is skipping the key idea: "Solve with inverse operations, then graph the solution set: open circle for $<$ or $>$, closed for $\le$ or $\ge$." — always check your work against this rule before you submit.*
8. **Try It 1:** A. Yes, because $5 \le 5$ is true — *Substitute $x = 5$ into $x \le 5$: $5 \le 5$ is true because $\le$ means 'less than OR equal to.' The boundary value 5 IS a solution, so the gate opens.*
9. **Try It 2:** A. All numbers greater than 7, like 8, 9, 10, … (but NOT 7 itself) — *The solution set is every value that makes $x > 7$ true. That means all numbers greater than 7, such as 8, 9, 10, and beyond. Because $>$ does not include the boundary, 7 itself is NOT in the solution set.*
10. **Exit Ticket:** A. $x < 6$; open circle at 6, shade left — *Subtract 5 from both sides: $x < 11 - 5 = 6$. Open circle at 6 (not included), shade left. ✓*